1. A spherical balloon is inflated with gas at the rate of 800 cubic centimeters per minute. How fast is the radius of the balloon increasing at the instant the radius is

(a) 30 centimeters

(b) 60 centimeters

2. A ladder 25 feet long is leaning against the wall of a house. The base of the ladder is pulled away from the way at a rate of 2 feet per second. How fast is the top of the ladder moving down the wall when its base is 7 feet from the wall? 15 feet from the wall?

3. The radius r of a sphere is increasing at a rate of 3 inches per minute.

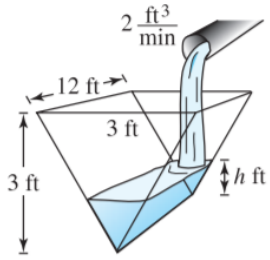
(a) Find the rates of change of the volume when $r = 9$ inches and $r = 36$ inches.

(b) Explain why the rate of change of the volume of the sphere is not constant even though $\dfrac{dr}{dt}$ is constant.

4. At a sand and gravel plant, sand is falling off a conveyor and onto a conical pile at a rate of 10 cubic feet per minute. The diameter of the base of the cone is approximately three times the altitude. At what rate is the height of the pile changing when the pile is 15 feet high? For reference, the volume of a cone is $V = \dfrac{1}{3}\pi r^2 h$.

5. A conical tank (with vertex down) is 10 feet across the top and 12 feet deep. If water is flowing into the tank at a rate of 10 cubic feet per minute, find the rate of change of the depth of the water when the water is 8 feet deep.

6. A trough is 12 feet long and 3 feet across the top. It's ends are isosceles triangles with altitudes of 3 feet. If water is being pumped into the trough at 2 cubic feet per minute, how fast is the water level rising when the depth h is 1 foot?

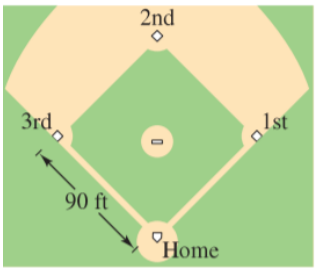


7. A ladder 25 feet long is leaning against the wall of a house (see figure). The base of the ladder is pulled away from the way at a rate of 2 feet per second.

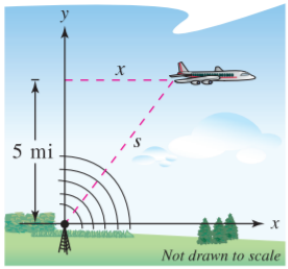
(a) Consider the triangle formed by the side of the house, the ladder, and the ground. Find the rate at which the area of the triangle is changing when the base of the ladder is 7 feet from the wall.

(b) Find the rate at which the angle between the ladder and the wall of the house is changing when the base of the ladder is 7 feet from the wall.

8. A baseball diamond has the shape of a square with sides 90 feet long. A player running from second base to third base at a speed of 25 feet per second is 20 feet from third base. At what rate is the player's distance s from home plate changing?



9. An airplane is flying at an altitude of 5 miles and passes directly over a radar antenna (see figure). When the plane is 10 miles away ($s = 10$)), the radar detects that the distance s is changing at a rate of 240 miles per hour. What is the speed of the plane?



10. A man 6 feet tall walks at a rate of 5 feet per second away from the light that is 15 feet above the ground. When he is 10 feet from the base of the light,

(a) at what rate is the tip of his shadow moving?

(b) at what rate is the length of his shadow changing?

11. A balloon rises at a rate of 4 meters per second from a point on the ground 50 meters from an observer. Find the rate of change of the angle of elevation of the balloon from the observer when the balloon is 50 meters above the ground.

12. An airplane flies at an altitude of 5 miles toward a point directly over an observer. The speed of the plane is 600 miles per hour. Find the rate at which the angle of elevation θ is changing when the angle $\theta = \pi/6$.

13. Two cars leave an intersection. One travels north at 30 miles per hour, the other travels east at 40 miles per hour. How fast is the distance between them increasing at the end of 30 minutes?